

Smart Waste Monitoring System

Analytics & Performance Report

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Generated By: Smart Waste Monitoring System

System Overview

This report presents analytics generated from the smart waste monitoring system. The system analyses bin fill levels, identifies collection priorities using a rule-based DSS, and logs collection events separately from sensor telemetry.

Metric	Value
Total Telemetry Records	510
Average Fill Level	46.5%
Critical Records (≥80%)	82
Highest Risk Bin	BIN-005
Highest Fill Level	94%
Total Collection Events	42
Most Collected Bin	BIN-002
Avg Fill at Collection	98.2%

Historical Trend Analysis

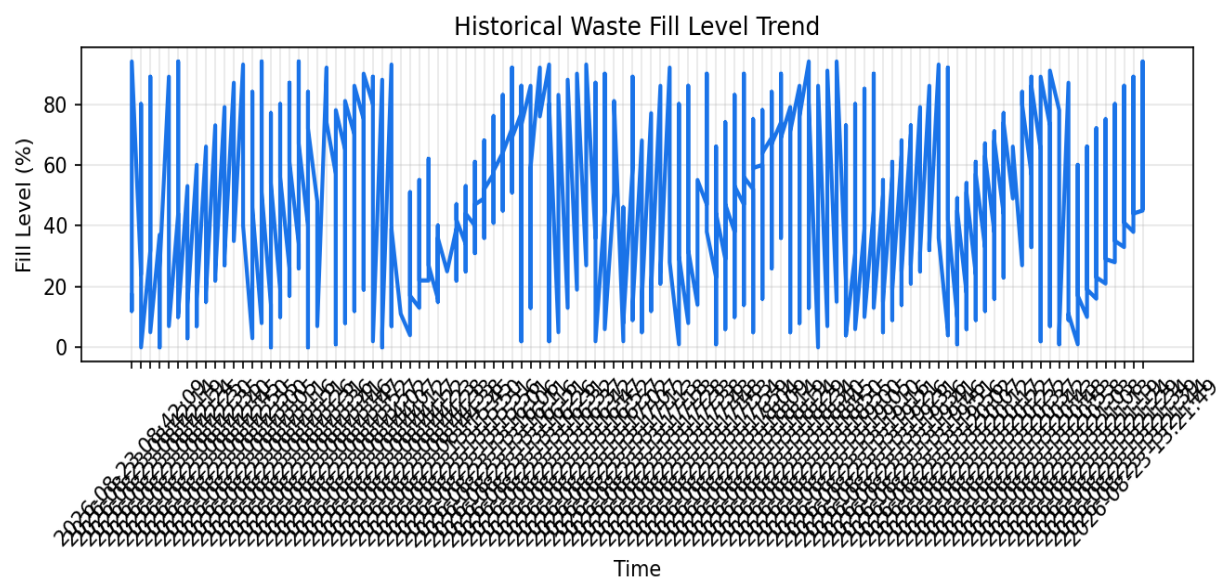


Figure 1. Historical waste fill level trend across recorded sensor readings.

Risk Distribution Analysis

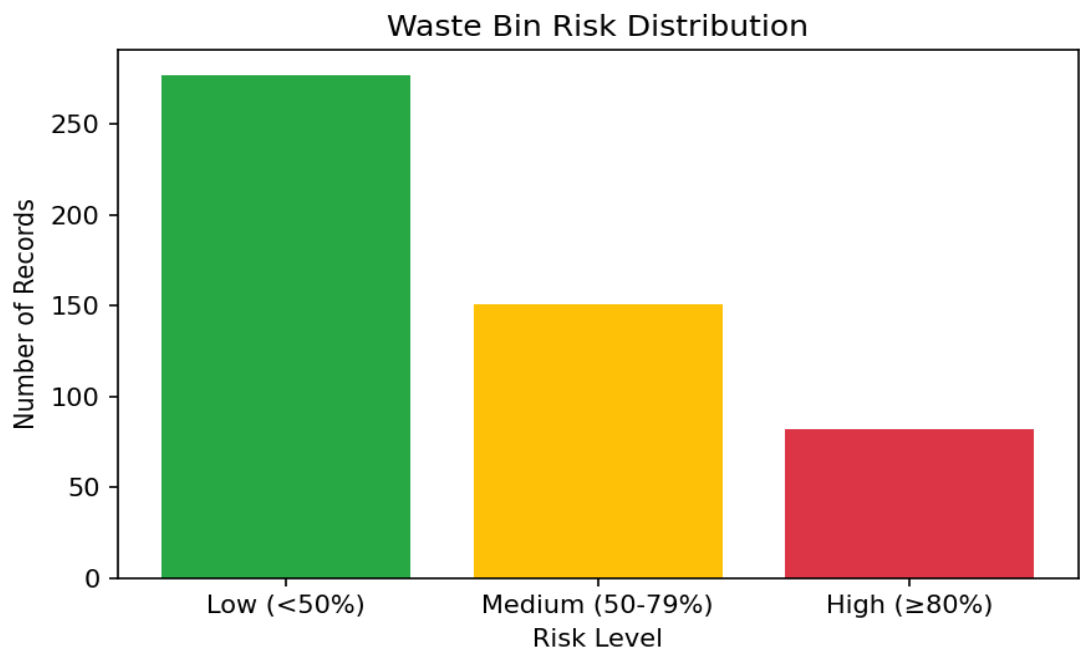


Figure 2. Distribution of low, medium, and high-risk waste records.

Collection Events Summary

The following table summarises bin collection events recorded in the *collection_events* table — stored separately from sensor telemetry to enable clean queryability of actual emptying events.

Bin ID	Location	Fill at Collection (%)	Collected At
BIN-001	Colombo Fort	95%	2026-08-23 13:21:13
BIN-002	Pettah Market	99%	2026-08-23 13:21:03
BIN-005	Wellawatte	97%	2026-08-23 13:21:03
BIN-004	Bambalapitiya	97%	2026-08-23 13:20:53
BIN-003	Slave Island	99%	2026-08-23 13:20:12
BIN-001	Colombo Fort	99%	2026-08-23 13:20:07
BIN-002	Pettah Market	100%	2026-08-23 13:20:07
BIN-004	Bambalapitiya	95%	2026-08-23 13:19:31
BIN-002	Pettah Market	99%	2026-08-23 13:19:10
BIN-005	Wellawatte	98%	2026-08-23 13:19:10

System Assessment

The analytics module successfully evaluates historical waste accumulation trends and identifies high-priority collection requirements. The rule-based DSS applies configurable fill-level thresholds to generate actionable recommendations. Collection events are stored in a dedicated table, enabling independent analysis of collection frequency and bin-level performance.